

Color Theory

Portfolio Design + Game UI



COURSE CSGG2012P · CO3
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PROGRAM BTech CSE · Graphics & Gaming · Sem IV
INSTITUTE UPES Dehradun
MODALITY Portfolio Website + Chroma Color Game
SCHEME Analogous — three greens on the color wheel



The Color Wheel

Foundation of all color relationships

PRIMARY COLORS

Cannot be created by mixing other colors.

● **Red**

● **Yellow**

● **Blue**

SECONDARY COLORS

Created by mixing two primaries.

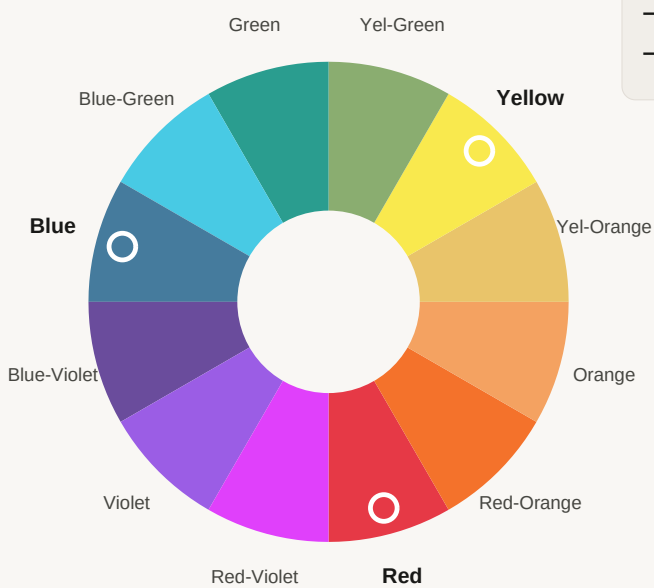
■ **Orange** = R + Y

■ **Green** = Y + B

■ **Violet** = B + R

KEY PRINCIPLE

- **Adjacent colors** → Analogous harmony
- **Opposite colors** → Complementary contrast
- **3 equidistant colors** → Triadic balance
- **One hue, varied lightness** → Monochromatic



Color Schemes

How colors combine to create visual harmony



ANALOGOUS

USED IN PROJECT

Colors adjacent on the wheel. Natural, harmonious feel — like greens in nature. Chosen for calm, creative atmosphere.

#4D6940 · #688B58 · #8AAD70

See scheme applied on page 5 & 6



COMPLEMENTARY

180° apart on the wheel. Maximum contrast, high visual energy. Used in Mission 03 of the Chroma game.

#E63946 ↔ #2A9D8F (opposite sides)

See scheme applied on page 5 & 6



TRIADIC

Three colors 120° apart. Vibrant and balanced. Strong contrast without losing cohesion or harmony.

Red · Yellow · Blue (120° spacing)

See scheme applied on page 5 & 6



MONOCHROMATIC

One hue — varied tints, shades and tones. Clean, cohesive, elegant. Used in Mission 04 of Chroma.

Forest → Sage (same hue, varied lightness)

See scheme applied on page 5 & 6

The RGB Model

How light combines to create color on digital screens

HOW LIGHT MIXES

Red + Green → Yellow
Red + Blue → Magenta
Green + Blue → Cyan
R + G + B → White
No light → Black

CHANNEL VALUES (0–255)

0 = none of that color · 255 = full intensity

PROJECT PRIMARY GREEN

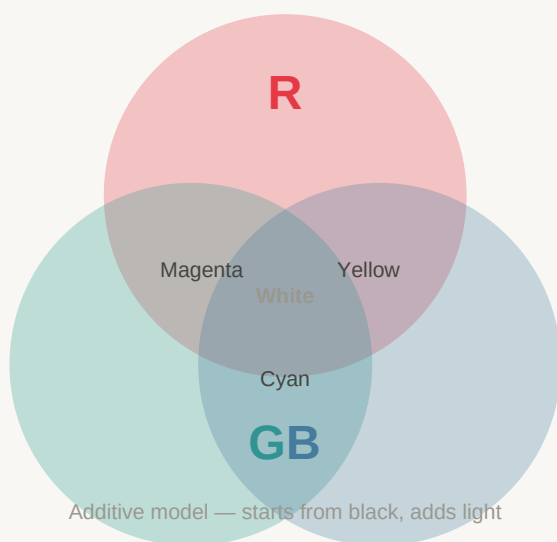


#688B58 = rgb(104, 139, 88)

R = 104

G = 139

B = 88



ADDITIVE VS SUBTRACTIVE

Additive (RGB)

Light on screen. Black start. R+G+B = White.

Subtractive (CMYK)

Ink on paper. White start. C+M+Y+K = Black.

Project Palette

Six colors — each with a defined role in the design system

ANALOGOUS SCHEME: COLUMNS 1–4 FORM GREEN PROGRESSION ON THE COLOR WHEEL



BACKGROUND

Ink Black

#031926 · rgb(3, 25, 38)

Deep navy base — the canvas for the entire site



PRIMARY

Dusty Olive

#688B58 · rgb(104, 139, 88)

Core brand color — headings, buttons, key UI



SECONDARY

Sage Green

#8AAD70 · rgb(138, 173, 112)

Lighter analogue — tags, highlights, secondary UI

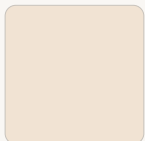


ACCENT

Mauve Bark

#775B59 · rgb(119, 91, 89)

Warm contrast — interactive elements, hover states

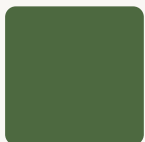


TEXT / FG

Almond Cream

#F1E3D3 · rgb(241, 227, 211)

All body text — warm off-white, high readability



DARK ANALOGUE

Forest Deep

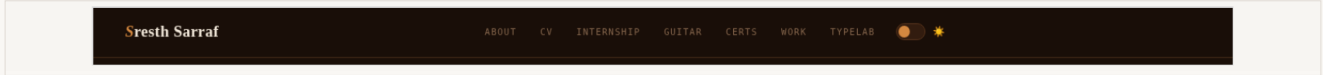
#4D6940 · rgb(77, 105, 64)

Darkest green — depth, shadows in game UI

Global Color Accent

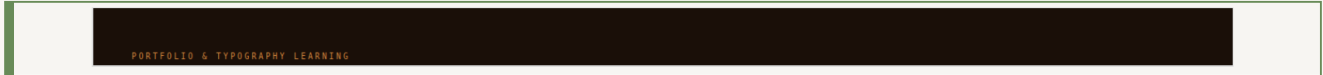
Amber → Olive Green · No scheme → Analogous scheme

BEFORE Accent: #D48840 Amber · Nav: About / CV / Internship / Guitar / Certs / Work / TypeLab



— Amber accent chosen without color theory framework · No "Game" link (section didn't exist)

AFTER Accent: #688B58 Dusty Olive · Nav: About / Internship / Guitar / ■ Game / Work / TypeLab



— Olive green replaces amber · "Game" link added · Accent now part of analogous scheme

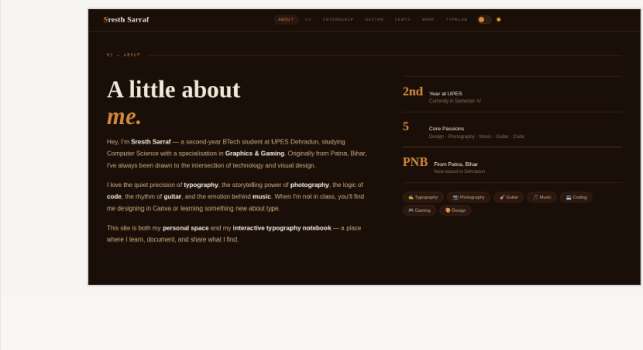
WHY THIS MATTERS

- The nav accent defines first impression — it had to align with the color scheme.
- Adding "■ Game" to nav made the Chroma game section discoverable.
- Olive (#688B58) is the Primary color in the analogous palette.

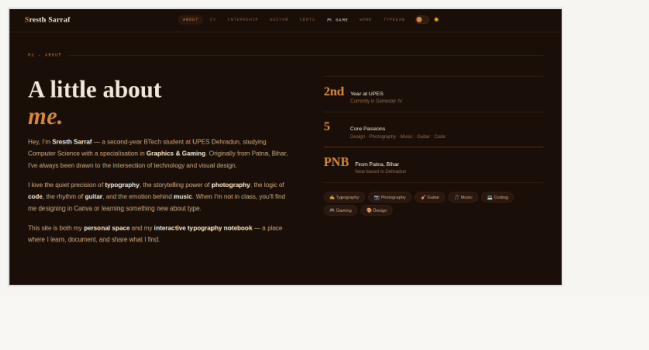
Hero Background & Typography Color

Brown #1A0F08 → Ink Black #031926 · Amber heading → Olive heading

BEFORE



AFTER



WHAT CHANGED

Element	Before	After
Background	#1A0F08 warm brown	#031926 ink black — deeper, more intentional
Italic heading	#D48840 amber — not part of any scheme	#8AAD70 sage green — secondary colour in palette
Nav highlight	Amber — warm but unstructured	Olive — now part of analogous triad

Color Theory Game Section

Built from scratch — teaches all 4 color schemes via interactive missions

BEFORE Slot occupied by Assignments — no color theory content **AFTER** New Chroma game — palette strip, missions, color role cards



WHAT THIS ADDS

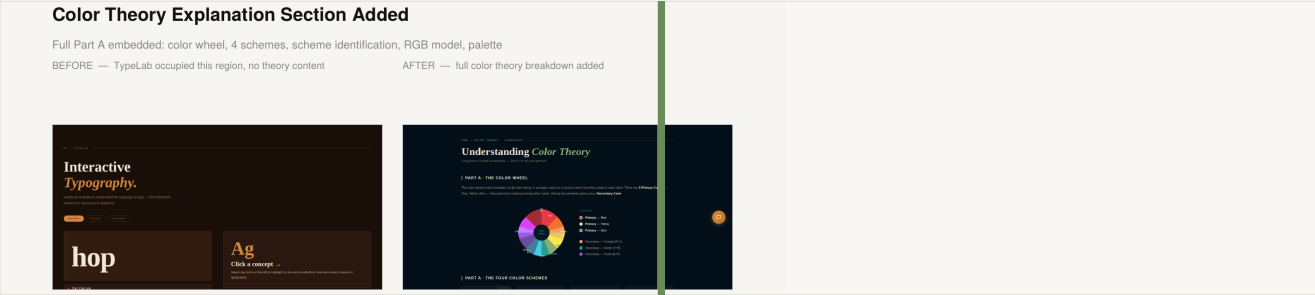
- Interactive palette strip showing all 6 project colors with HEX + RGB values
- Mission 01–04 covering Analogous, Complementary, Triadic, Monochromatic
- Color role cards defining Primary, Secondary, Accent, Background in the system
- Direct link to the browser-based Chroma game — playable from the portfolio

CHANGE 04

Color Theory Explanation Section Added

Full Part A embedded: color wheel, 4 schemes, scheme identification, RGB model, palette

BEFORE — TypeLab occupied this region, no theory content AFTER — full color theory breakdown added

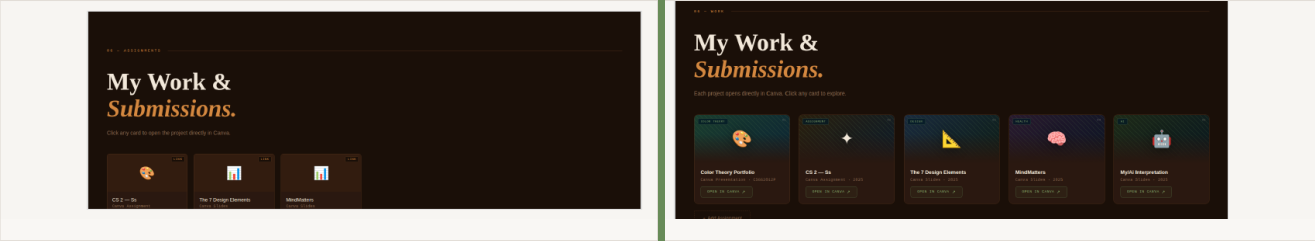


CHANGE 05

Work Section — Cards Redesigned + New Entry Added

Generic flat cards → gradient cards with icons, type tags, number badges. Color Theory Portfolio added.

BEFORE — 4 flat brown cards, no icons, generic "link" badge AFTER — 5 cards with gradients, icons, type tags, Color Theory entry



Summary

- 01 **Navigation Accent**
Amber → Olive. Accent now part of analogous scheme.
- 02 **Hero Section**
Brown bg → Ink Black. Heading accent → Sage.
- 03 **Chroma Game**
Entirely new section. Teaches all 4 schemes.
- 04 **Theory Section**
Full Part A explanation embedded in site.
- 05 **Assignments**
Colors updated to match analogous palette.

PROJECT PALETTE

Analogous scheme: 3 greens on the color wheel



Chroma game: 4 color theory missions

CSGG2012P · 40 marks · CO3

PART A — COLOR THEORY (20 MARKS)

COLOR WHEEL

Diagram — 12 segments, Primary & Secondary labeled

FOUR SCHEMES

Analogous, Complementary, Triadic, Monochromatic

IDENTIFICATION

Analogous chosen — three greens side-by-side on wheel

RGB MODEL

Venn + mixing rules + connection to project palette

PART B — SCHEME APPLIED (20 MARKS)

SCHEME

Analogous — 3 greens (#4D6940 / #688B58 / #8AAD70)

PALETTE

6 colors with HEX + RGB + defined roles

MODALITY

Portfolio site + Chroma Color Game UI

BEFORE / AFTER

5 documented changes across site sections

DECLARATION

I, Sresth Sarraf, have applied color theory principles to my portfolio website and Chroma game UI as part of CSGG2012P CO3. The analogous scheme was chosen to reflect a calm, nature-inspired design language.